Physical System Specification and Operational Architecture

RIS-Aided Secure Sensing System

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Outline

System Overview

Objective: Detailed specification of hardware, environment, and protocols for an Intelligent Reflecting Surface (RIS) aided secure sensing system.

Standards Alignment:

- Strictly aligned with **3GPP Release 16/17 standards**.
- Focus on 5G mmWave communications.
- Tailored for Industrial IoT (IIoT) applications.

The Smart Factory Environment (InF)

Based on **3GPP TR 38.901: InF-DH (Dense Clutter, High Base Station)**.

Characteristics

- High density of metallic machinery, storage racks, and mobile robots.
- **Propagation Challenge:** Severe Non-Line-of-Sight (NLoS) regions ("Hard Shadows").

Justification for RIS

At mmWave frequencies (> 30 GHz), diffraction is negligible. An RIS is required to create a **Virtual Line-of-Sight (vLoS)** bridge.

Component 1: The Transmitter (Source)

Type: 5G New Radio (NR) FR2 Small Cell (gNodeB).

- **Frequency: 32.8 GHz** (Ka-Band).
- **Antenna:** Directional Horn Antenna Array (Fixed orientation).

Operational Roles (3GPP TS 38.214):

1. **Phase 1 (Sensing):**
   - Transmits **CSI-RS** (unmodulated pilot tones).
   - Used for channel estimation/localization.
2. **Phase 2 (Communication):**
   - Transmits **PDSCH** (Physical Downlink Shared Channel).
   - Modulated high-bandwidth user data (QAM/OFDM).

Component 2: The RIS

The core passive relay element designed to bypass obstacles.

Hardware Specifications:

- **Array:** 8×8 Element Metasurface.
- **Quantization:** 1-Bit Phase (0 or π).
- **Amplitude Loss:** State-dependent efficiency.
  - $\alpha_{OFF} = 0.9$
  - $\alpha_{ON} = 0.4$

Operational Modes:

- **Phase 1 (Sweep):** Rapid switching (32-64 codebook patterns).
- **Phase 2 (Lock):** Frozen phase configuration for high-gain pencil beam.

Components 3 & 4: Obstacle & User

3. The Obstacle (Blocker)

- **Type:** Industrial CNC Machine / Metallic Cabinet (2m $\times$ 1m).
- **Material:** Perfect Electrical Conductor (PEC).
- **Effect:** Creates "Hard Shadow" ($-\infty$ dB gain).

4. The User (Receiver)

- **Type:** Autonomous Mobile Robot (AMR) / AGV.
- **Mobility:** Low velocity (≈ 1 m/s).
- **Function:** Measures **RSRP** (Signal Power) for localization; Demodulates 4K Video/LIDAR data.

Beam Management Protocol

Strictly sequential two-phase protocol dictated by mmWave propagation.

1. **Phase 1: Sensing (Beam Training)**
   - *Objective:* User Localization.
   - gNodeB sends CW pilots; RIS sweeps codebook.
   - Robot measures RSRP vector.
   - **Model:** Random Forest infers angular position.
2. **Phase 2: Communication (Data Tx)**
   - *Objective:* High-Bandwidth Data Link.
   - RIS forms static relay beam towards estimated angle.
   - Transfer of mission-critical data (e.g., 64-QAM).

Physically Calibrated Offline Model

To enable training prior to hardware deployment, we synchronize the software environment with physical reality.

Purpose of Simulation

- **De-Risking Fabrication:** Incorporates realistic losses (e.g., Amplitude Dip) to test algorithm robustness.
- **Pre-Emptive Debugging:** Identified "Symmetric Ambiguity" (Ghost Beams) in 1-Bit quantization.
- **Data Generation:** Generates 5,000+ samples for Random Forest training (infeasible to collect manually).

Info Sync (Physical-to-Digital)

Explicit parameter transfers ensuring the model is not merely a theoretical abstraction.

Parameter	**Physical Source**	**Algorithm**
Resonant Freq.	CST Microwave Studio	`wavelength = 3e8 / 32.8e9`
Reflection Loss	Skyworks SMP1345	`w_amps = 0.4(ON)/0.9(OFF)`
Quantization	1-Bit PIN Diode	`np.where(phase > pi/2, pi, 0)`
User Behavior	3GPP InF Stats	`np.random.uniform(0, 60)`

Table: Mapping Hardware Constraints to Python Code